

STANDARD OPERATING PROCEDURE

SOP-CDU2-014 Rev 4 | Pump Changeover - Hot Standby, Column Bottoms Service

Document No.	SOP-CDU2-014	Revision	Rev 4
Effective Date	2026-02-15	Next Review	2028-02-14
Unit	CDU-2 - Crude Distillation Unit - 2	Document Owner	A. Fernandes
Equipment	P-2104A / P-2104B (Column Bottoms RCO Pumps)	Classification	Safety Critical

1. PURPOSE

This procedure defines the mandatory sequence for transferring duty between the two CDU-2 column bottoms pumps, P-2104A and P-2104B, while the unit remains on line. The service is Reduced Crude Oil (RCO) at a normal operating temperature of 345 degC, which is above the auto-ignition temperature of the fluid. Any loss of containment during changeover is therefore treated as a potential fire event.

2. SCOPE AND APPLICABILITY

Applicable to all CDU-2 shift personnel executing a planned or unplanned hot standby changeover of P-2104A/B. This procedure does not cover pump isolation for maintenance, which is addressed by the unit isolation and positive-isolation permit procedure.

3. RESPONSIBILITIES

Role	Responsibility
Shift In-Charge	Authorises the changeover. Must hold a current 'Pump Changeover - Hot Standby' certification. Verifies all pre-start checks in Section 5 are signed before authorising.
Panel Operator	Monitors column bottoms level on C-2101, discharge pressure and flow throughout. Acknowledges and records all alarms.
Field Operator	Executes valve operations in the sequence given in Section 6. Confirms each step verbally to the Panel Operator.
Rotating Equipment Engineer	Consulted where the pump being taken off line has any condition monitoring exceedance. Approves return to service after any corrective work.

4. VIBRATION AND TEMPERATURE LIMITS

Condition monitoring limits for P-2104A/B are set per ISO 10816-3, Group 1, rigid mounting. These limits are mandatory decision points, not advisory values.

Parameter	Normal	Alert	Trip / Remove from service
Overall vibration velocity (mm/s RMS)	up to 4.5	7.1	11.0
Bearing temperature (degC)	up to 75	85	95
Seal chamber leakage	Nil	Visible weeping	Continuous drip or mist

A reading at or above the alert value of 7.1 mm/s RMS requires a corrective work order to be raised within 24 hours and the standby pump to be confirmed available. A reading at or above the trip value of 11.0 mm/s RMS requires changeover at the earliest safe opportunity.

5. PRE-START CHECKS ON THE INCOMING PUMP

- 5.1 Confirm the incoming pump casing is warm. Warm-up line to be cracked open for a minimum of 30 minutes before start. Casing temperature to be within 40 degC of the discharge header temperature.
- 5.2 Confirm suction valve is FULLY OPEN and the discharge valve is CLOSED.
- 5.3 Confirm the non-return valve (NRV-2104A or NRV-2104B as applicable) is in service and has not been isolated.

- 5.4 Confirm bearing lubricant level is between the sight glass marks and the oil is clear with no water haze.
- 5.5 Confirm the mechanical seal support system is healthy: barrier or buffer fluid level above minimum, no low level alarm standing. RECORD THE ACTUAL LEVEL, do not simply tick.
- 5.6 Confirm laser alignment record exists and shows parallel offset not exceeding 0.05 mm, if the pump has been opened since last run.
- 5.7 Confirm cooling water flow to the seal cooler is established and the return is warm to touch.
- 5.8 Confirm the pump is selected to AUTO on the DCS and the auto-start permissive is healthy.
- 5.9 Confirm the area is free of hot work and that no permit is active within 15 m. Refer to any standing safety circular for the circuit.

6. CHANGEOVER SEQUENCE

Steps must be executed in the order given. Each step is confirmed verbally between the Field Operator and the Panel Operator before the next step begins. If any step cannot be completed as written, STOP and revert to the original line-up.

- 6.1 Shift In-Charge authorises the changeover and records the reason and time in the shift log.
- 6.2 Panel Operator notes the current C-2101 bottoms level, bottoms flow and discharge pressure as the reference condition. Place the level controller LIC-2101 on MANUAL at the current output.
- 6.3 Panel Operator raises the C-2101 bottoms level to the upper half of the normal band to provide inventory margin during the transfer.
- 6.4 Start the incoming pump. Confirm discharge pressure develops to within 0.5 kg/cm² of the running pump within 10 seconds. If pressure does not develop, STOP the pump immediately and investigate before any further attempt.
- 6.5 Crack open the incoming pump discharge valve slowly, taking not less than 60 seconds to reach 25% open. Watch for discharge pressure interaction and for any change in motor current on both machines.
- 6.6 Continue opening the incoming discharge valve to fully open while simultaneously throttling the outgoing pump discharge valve, maintaining total bottoms flow within 5% of the reference value throughout.
- 6.7 When the incoming pump is carrying full flow, close the outgoing pump discharge valve fully.
- 6.8 Stop the outgoing pump. Confirm it comes to rest without reverse rotation. If reverse rotation is observed, report immediately as a suspect non-return valve.
- 6.9 Return LIC-2101 to AUTO once the level is stable for 10 minutes.
- 6.10 Line up the stopped pump as the new standby: discharge valve closed, suction open, warm-up line cracked open, AUTO selected. Unless the pump is being handed over for maintenance, in which case follow the isolation procedure and do NOT select AUTO.
- 6.11 Record in the shift log: time of changeover, reason, which pump is now lead, final vibration and bearing temperature readings on the running pump, and the name of the authorising Shift In-Charge.

7. ABNORMAL CONDITIONS

Condition observed	Immediate action
Incoming pump does not develop discharge pressure	Stop the pump. Do not re-attempt. Suspect vapour lock or a closed suction. Verify warm-up and suction line-up. Inform Shift In-Charge.
Bottoms level falls below the low alarm during transfer	Halt valve movement. Restore flow on whichever pump is developing pressure. Reduce heater firing on F-2101 if the level continues to fall.
Visible hydrocarbon release at either seal	Declare an emergency per the unit emergency response procedure. Stop the leaking pump only if the other pump is confirmed carrying flow. Activate area gas detection acknowledgement and clear non-essential personnel from the pump house.
Vibration on the incoming pump exceeds 7.1 mm/s after start	Do not transfer load. Restore the original line-up. Raise a corrective work order and inform the Rotating Equipment Engineer.
Both pumps unavailable	Reduce F-2101 firing and cut unit rate immediately. Bottoms level high-high will trip the heater; anticipate and control rather than allow the trip.

8. REFERENCED DOCUMENTS

Document	Title
ENG-STD-SUMM-119	Internal Engineering Summary - Hot Service Pump Requirements
HAZOP-CDU2-2024	Column Bottoms Circuit - Action Tracking Register (Node 7)
VIB-CDU2-2026	CDU-2 Rotating Equipment Vibration and Bearing Temperature Log
SC-2026-11	Restriction on Hot Work - CDU-2 Column Bottoms Circuit
PID-CDU2-003	CDU-2 Column Bottoms Circuit - Simplified Flow Schematic

9. REVISION HISTORY

Rev	Date	Change	Approved by
1	2019-05-10	First issue	Head of Operations
2	2021-08-22	Added seal support system pre-start check	Head of Operations
3	2024-01-30	Vibration limits aligned to ISO 10816-3; abnormal conditions table added	MRPL Ops
4	2026-02-15	Warm-up duration increased to 30 minutes; discharge valve opening rate specified; hot work check added to pre-start	A. Fernandes
5 (draft)	under review	Pending: mandatory laser alignment record with 0.05 mm tolerance before return to service. Arising from CAPA-0042-2 of INC-2026-0042.	Not yet approved